

HOW ACCURATE IS GOOD ENOUGH?

Approximating a Civil Engineering Function Using Infinite Series

An Integrated Python Exercise for Civil Engineering Students

Topics Covered:

- Geometric Series
 - Power Series
- Maclaurin Series
 - Taylor Series
- Error Analysis and Engineering Decision-Making

Learning Outcomes:

Students will understand how infinite series can approximate engineering functions, quantify approximation errors, and make engineering decisions based on accuracy requirements.

VILLONDO, Angel Rain L.

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ENGINEERING SCENARIO

A civil engineer is analyzing the geometry of a sloping structural or surveying system. The relationship between angle and vertical component is given by:

$$y = L \sin(\theta)$$

Where:

- $L = 20$ m (structural length)
- θ = angle in radians
- y = vertical component

Central Question: How accurately can a civil engineer approximate an engineering quantity using a finite number of terms of an infinite series?

ANSWER:

A civil engineer can approximate quantities with high precision provided that the number of terms is sufficient for the target angle or value and that the evaluation point remains close to the expansion center. Using finite terms introduces a truncation error, which must be quantified to ensure it stays below engineering error tolerances (such as $< 0.1\%$).

PART 1: GEOMETRIC SERIES

Begin with the geometric series and its partial sum:

$$1 + x + x^2 + x^3 + \dots = \sum x^k$$

Task: Implement a Python function to calculate the partial sum without using the closed-form equation $1/(1-x)$.

```
def geometric_sum(x, N): """Calculate S_N = 1 + x + x^2 + ... + x^N""" total = 0.0 for k in range(N + 1): total += x ** k return total
```

Investigation: Compare partial sums with the exact value for $x = 0.5$, 0.8 , and 0.9 . How does the number of terms affect accuracy? Why does convergence speed differ?

Key Concepts: partial sum, convergence, approximation error, number of terms needed

ANSWER:

How the number of terms affects accuracy: As more terms (N) are added to the partial sum $1 + x + x^2 + \dots + x^N$, the approximation gets closer to the closed-form exact value of $1/(1-x)$.

Why convergence speed differs: Convergence speed depends directly on the magnitude of x . For $x = 0.5$, powers of x shrink very rapidly ($0.5^N \rightarrow 0$), requiring fewer terms for convergence. For $x = 0.8$ and 0.9 , powers of x shrink much more slowly, meaning a significantly larger number of terms is required to achieve the same level of accuracy.

PART 2: POWER SERIES

Generalize the geometric series into a power series:

$$P_N(x) = a_0 + a_1x + a_2x^2 + \dots + a_Nx^N$$

Task: Create a function that evaluates a power series at a given point.

```
def power_series(x, coefficients): """Evaluate P_N(x) with given coefficients""" result
= 0.0 for k, a k in enumerate(coefficients): result += a k * (x ** k) return result
```

Key Insight: A polynomial is a finite power series. An infinite power series can approximate transcendental functions like sin, cos, and exponentials.

PART 3: MACLAURIN SERIES FOR $\sin(\theta)$

The Maclaurin series is a power series centered at $\theta = 0$:

$$\sin(\theta) = \theta - \theta^3/3! + \theta^5/5! - \theta^7/7! + \dots$$

Task: Build the Maclaurin approximation explicitly using loops. Do NOT use `math.sin()` to generate the approximation (only for verification).

```
def sin_maclaurin(theta, N): """Approximate sin(theta) with N terms""" result = 0.0 for n in range(N): sign = (-1) ** n factorial = math.factorial(2*n + 1) result += sign * (theta ** (2*n + 1)) / factorial return result
```

Investigation: For $\theta = 10^\circ$, calculate approximations using 1, 2, 3, and 4 terms. Compare each to the exact value. What is the error at each step?

ANSWER:

1-Term Approximation ($\theta - (\theta^3)/3!$ using 1 term $\rightarrow \theta$): Yields an approximation based purely on the linear small-angle assumption, with a higher percentage error because higher-order curvature is omitted.

2-Term Approximation: Adds the $-(\theta^3)/6$ term, drastically reducing the error.

3-Term Approximation: Adds the $+(\theta^5)/5!$ term, reducing the error further into very high-precision bounds.

4-Term Approximation: Includes up to the $(\theta^7)/7!$ term, achieving near-exact agreement with the true value of $\sin(10^\circ)$ where absolute and percentage errors drop to minimal fractions.

PART 4: ENGINEERING INVESTIGATION

Engineering Problem: Calculate the vertical component $y = L \sin(\theta)$ for various angles using Maclaurin approximations.

Given Data:

- $L = 20 \text{ m}$
- Angles: $1^\circ, 2^\circ, 5^\circ, 10^\circ, 15^\circ, 20^\circ, 30^\circ$
- Use Maclaurin approximations with 1, 2, 3, and 4 terms

Task: Create a table for each number of terms showing:

- Angle (degrees)
- Exact value y
- Approximation
- Absolute error
- Percentage error

Analysis Questions:

1. How does the error change as the angle increases?
 - The approximation error increases exponentially as the angle θ increases away from the expansion center (0° for Maclaurin series). Higher angles require a larger domain expansion, causing higher-order terms to grow rapidly.
2. How does the error change as you add more terms?
 - Adding more terms systematically decreases the approximation error, bringing the polynomial approximation closer to the exact trigonometric function value. Convergence is fast initially but slows down at wider angles unless additional terms are included.
3. For small angles ($< 5^\circ$), how many terms are sufficient?
 - For small angles under 5° , a 1-term approximation ($\sin(\theta) \approx \theta$) is fully sufficient to satisfy standard engineering error bounds such as the $< 0.1\%$ error tolerance.
4. For larger angles ($> 20^\circ$), how many terms are needed?
 - For larger angles exceeding 20° (up to 30°), at least 4 terms (or a shifted Taylor series expansion) are required to maintain high precision and keep the percentage error under 0.1% .

PART 5: TAYLOR SERIES

The Taylor series generalizes from Maclaurin by expanding around any point a :

$$f(\theta) = f(a) + f'(a)(\theta - a) + f''(a)/2!(\theta - a)^2 + \dots$$

Key Insight: Maclaurin is simply Taylor with $a = 0$. By choosing a different expansion point, you can improve accuracy in different regions.

Task: Construct a Taylor approximation for $\sin(\theta)$ centered at $a = 10^\circ$.

```
def sin_taylor(theta, a, N): """Approximate sin(theta) using Taylor centered at a"""
    result = 0.0
    sin_a, cos_a = math.sin(a), math.cos(a)
    for n in range(N): # Derivatives of sin cycle every 4 terms
        derivative_pattern = n % 4
        if derivative_pattern == 0: f_deriv = sin_a
        elif derivative_pattern == 1: f_deriv = cos_a
        elif derivative_pattern == 2: f_deriv = -sin_a
        else: f_deriv = -cos_a
        term = f_deriv * ((theta - a) ** n) / math.factorial(n)
        result += term
    return result
```

Comparison: For the same angles (1° – 30°), calculate errors using:

- Maclaurin (centered at 0°)
- Taylor (centered at 10°)

Which performs better near 10° ? Which is better far from 10° ?

ANSWER:

The Taylor series centered at 10° performs significantly better near 10° [cite: 1]. Because its expansion point is located right at 10° , the distance term $(\theta - a)$ is minimized, resulting in a much smaller truncation error in that local region.

The Maclaurin series (centered at 0°) performs better far from 10° in the direction of the origin (near 0°), because Taylor series accuracy degrades rapidly the further away an evaluation point moves from its chosen expansion center. As shown in the error comparisons, a series loses its precision as the distance $|\theta - a|$ increases.

PART 6: ENGINEERING DECISION—ERROR TOLERANCE

Requirement: The approximation must have less than 0.1% error.

Task 1: For each angle (1° – 30°), find the minimum number of terms needed to meet the 0.1% tolerance using both Maclaurin and Taylor approximations.

Task 2: Investigate the small-angle approximation $\sin(\theta) \approx \theta$. At what angle does this 1-term approximation stop being sufficiently accurate?

Analysis:

1. Does adding more terms always improve the approximation?
 - Within the radius of convergence for Maclaurin and Taylor series, yes. However, computationally, adding excessive terms past machine precision can run into round-off limitations, though theoretically they reduce truncation error.
2. Why is the Maclaurin approximation most accurate near zero?
 - Maclaurin series are power series centered specifically at $\theta = 0^\circ$. Because all higher-order derivative matching occurs at this single pivot point, the polynomial shares maximum local curve alignment right around zero, degrading with distance.
3. Why can changing the Taylor expansion point improve accuracy?
 - Shifting the expansion center closer to the region of interest (e.g., centering at 10° or 20° instead of 0°) minimizes the distance term $(\theta - a)$, drastically reducing truncation errors for measurements occurring away from the origin.
4. How does the distance from the expansion point affect accuracy?
 - Accuracy is inversely proportional to distance; the further an evaluation point is from the chosen expansion center a , the larger the approximation error becomes.
5. What is the critical angle for the $\sin(\theta) \approx \theta$ approximation?
 - The critical angle where the single-term linear approximation stops satisfying the strict 0.1% error tolerance threshold is approximately 5° . Beyond this limit, higher-order curvature terms ($-(\theta^3)/3!$) become mathematically significant.

PART 7: FINAL ENGINEERING RECOMMENDATION

Decision Problem:

You are the civil engineer responsible for the calculation. You need an approximation with less than 0.1% error. Would you use the Maclaurin series, a Taylor series centered closer to the angle of interest, or the exact sine function? Defend your choice using your numerical results.

Your recommendation must include:

- Number of terms required
- Percentage error achieved
- Convergence behavior
- Computational simplicity vs. accuracy tradeoff
- Valid angle range for your solution

REQUIRED PYTHON OUTPUT

Your final Python program must produce the following deliverables:

1. Numerical Tables

Formatted tables showing exact values, approximations, and errors for each angle and number of terms.

2. Convergence Plot

A figure showing how percentage error decreases as you add more terms, for different angles.

3. Function Comparison Plot

Side-by-side plots of exact $\sin(\theta)$ vs. Maclaurin and Taylor approximations with different numbers of terms.

4. Error Comparison Plot

Visual comparison of absolute and percentage errors between Maclaurin and Taylor series.

5. Written Recommendation

A clear explanation of your engineering decision with supporting numerical evidence.

Implementation Note: Explicitly implement the series summation using Python loops. Do not hide the mathematics inside symbolic mathematics software. The purpose is to make you see the progression from geometric series → power series → Maclaurin → Taylor → engineering approximation → error analysis → engineering decision.

THE MATHEMATICAL PROGRESSION

1. Geometric Series

$$1 + x + x^2 + \dots = 1/(1-x)$$



2. Power Series

$$a_0 + a_1 x + a_2 x^2 + \dots$$



3. Maclaurin Series

Expand around $\theta = 0$



4. Taylor Series

Expand around any point $\theta = a$



5. Engineering Approximation

Apply to $y = L \sin(\theta)$



6. Error Analysis

Compare with exact values



7. Engineering Decision

What's the best tradeoff?